Marketing Analytics

Segmentation and Customer Lifetime Value

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Session 06

A Retention-Support Decision

- A service team can support only some customers.
- High baseline value and responsiveness to support are separate quantities.
- Segments should correspond to feasible service decisions.

Customer Features at a Common Date

- Recency is time since the last eligible event.
- Frequency counts events in a fixed window.
- Monetary value uses sales or contribution over that window.

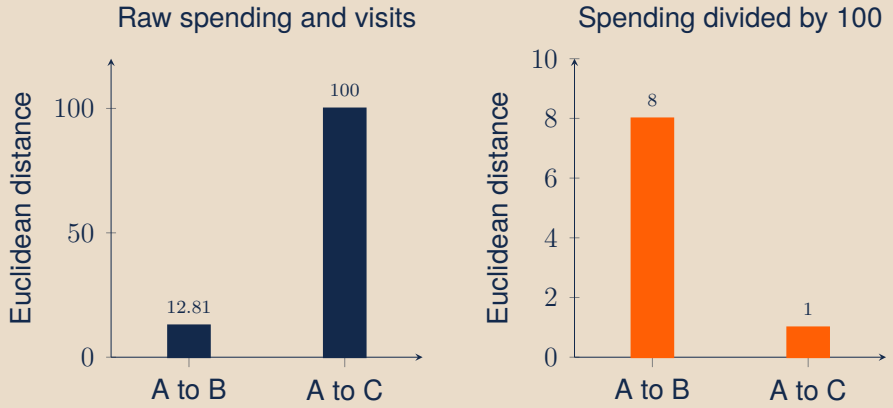
K-Means and Similarity

Choose K centers for transformed customer features z_i.

$$\min_{\mu, c} \sum_i \left\| z_i - \mu_{c(i)} \right\|^2$$

Scaling the features changes the distance and can change the segments.

Units Change the Nearest Customer



Synthetic customers have unchanged spending and frequency records.

Evaluating a Segmentation

- Inspect segment size, profile, and actionable differences.
- Compare stability across seeds, samples, and periods.
- Evaluate outcomes that did not mechanically define the clusters.

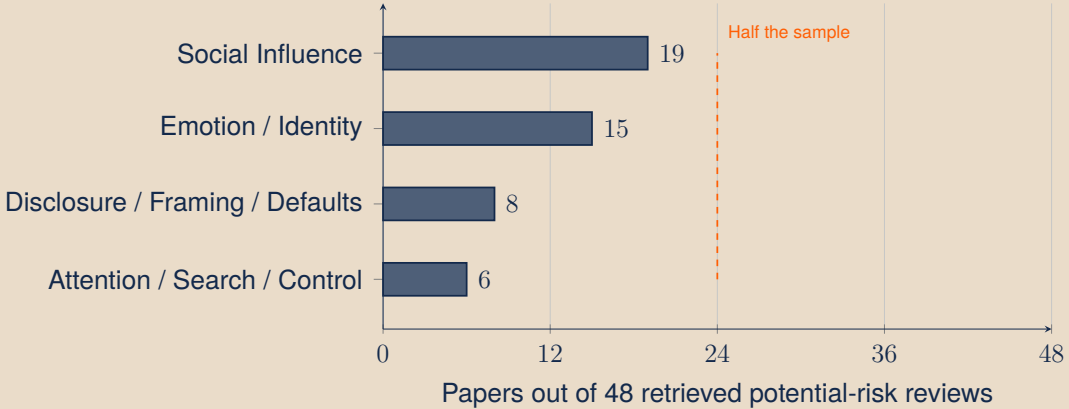
Similarity of Positive Mechanisms

For each paper, collect the mechanisms coded as present in a set A_i.

$$d_J(i,j) = 1 - \frac{|A_i \cap A_j|}{|A_i \cup A_j|}$$

Jaccard distance ignores shared absences. Average linkage compares these distances.

Social Influence is the Largest Cluster



48 retrieved potential-risk reviews. Four clusters selected from K=2 through K=8.

Reproduce the Mechanism Grouping

Run from the Consumer_Agency_Literature folder after installing its requirements.

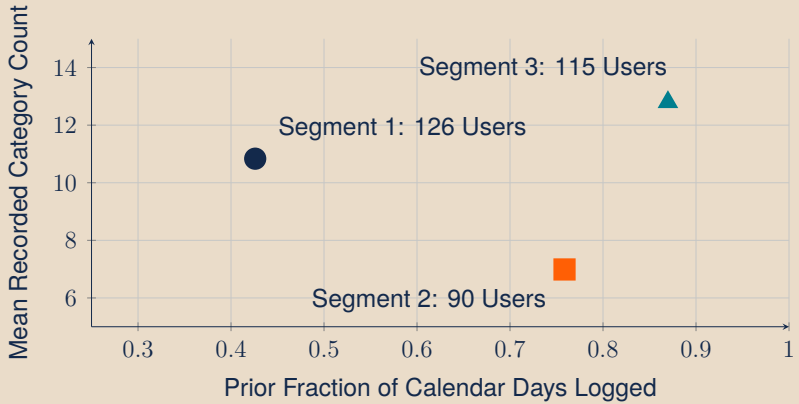
```
import pandas as pd
from cluster import FEATURES, select_sample, partition
df = pd.read_csv("coded_papers.csv")
sample = select_sample(df, "downloaded_risk")
labels, distances = partition(sample[FEATURES].to_numpy(), 4)
assert len(sample) == 48
```

The complete script compares K values, exports profiles, and refits sensitivity samples.

The Coding Changes the Conclusion

- Median subsample agreement: adjusted Rand index 0.794.
- Changing one feature per paper lowers median agreement to 0.250.
- The broader abstract-based sample selects eight clusters.

mHealth Segments Use Earlier Histories



Eligible users have at least 30 observed days before July 1.

Later Logging by Earlier Segment

Segment	Users	Prior Fraction	Later Fraction
1	126	0.426	0.338
2	90	0.759	0.594
3	115	0.870	0.654

mHealth denominators use 182 earlier and 184 later calendar days.

Lifetime Value and Payment Timing

Value future contribution conditional on remaining active.

$$\mathrm{CLV} = \sum_{t=1}^{T} \frac{\Pr(\text{active at } t) m_t}{(1+d)^t} - a$$

The period for margin m_t, discounting d, and retention must agree.

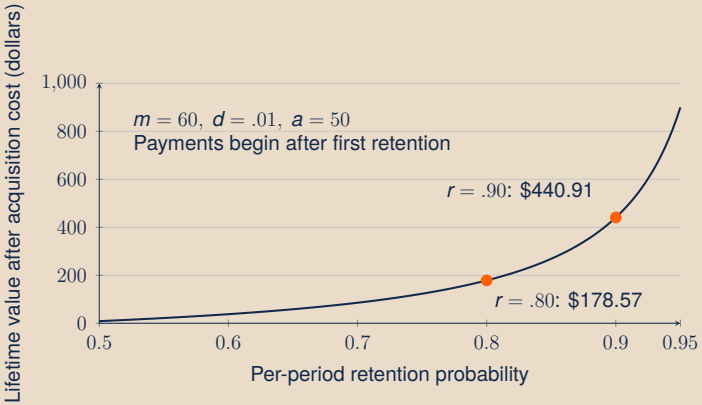
A Stationary Subscription

Suppose monthly retention is r and monthly contribution is m.

$$\mathrm{CLV} = \frac{mr}{1 + d - r} - a$$

With $m = 60$, $d = 0.01$, $a = 50$, and $r = 0.80$, CLV is 178.57 dollars.

Retention Has Nonlinear Value



Synthetic stationary model. Monthly margin, discounting, and acquisition cost stay fixed.

A Finite-Sum Check

Compare the closed form with 2,000 future monthly payments.

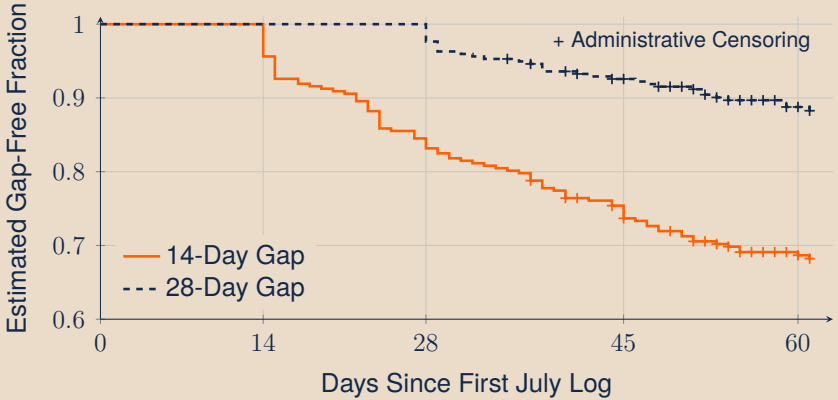
```
m, r, d, a = 60, 0.8, 0.01, 50
closed = m * r / (1 + d - r) - a
finite = sum(m * r**t / (1+d)**t
             for t in range(1, 2001)) - a
assert abs(closed - finite) < 1e-8
```

Both calculations give approximately 178.57 dollars.

Cancellation and a Logging Gap

- A subscription may record cancellation directly.
- A 30-day logging gap is an operational activity event.
- Observation can end before a defined event occurs.

The Event Definition Changes Survival



mHealth time-to-gap analysis uses a stated gap definition and observation horizon.

Endpoint and Continuous Retention

- Endpoint retention allows customers to leave and return.
- Continuous retention requires uninterrupted presence.
- Acquisition can include first entry, return, or switching.

Coding Exercise: Segments and Value

- Fit feature scaling and segments on an earlier period.
- Describe a later period without refitting the groups.
- Vary assumed paid retention in a separate CLV calculation.

Retention and Effective Cancellation

- High paid retention can reflect satisfaction or difficulty leaving.
- Measure cancellation completion and unwanted subsequent charges.
- Evaluate customer value alongside the seller’s lifetime-value objective.

Retention Discussion

- What action differs across the proposed segments?
- Does a customer’s absence represent churn or censoring?
- Which retention assumption changes the support decision?